Artificial Intelligence Technology and Application

Python
Lab Guide

Student Version

HUAWEI

HUAWEI TECHNOLOGIES CO., LTD.

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1 I/O Operation Experiment

1.1 Procedure

1.1.1 I/O Operations

Step 1 Write data to a file.

```
f = open("text.txt", 'w')          # Open the text.txt file. If the file does not exist, a new file will be
```

Output

```
Please enter the content to be written:
python operation experiment
```

Figure 1-1 Content to be written

```
python operation experiment
```

Figure 1-2 File content

Step 2 Read file data.

```
f = open("text.txt", 'r')
# Read six characters and move the cursor six characters backward.

# Read from the current position of the cursor to the end.
```

Output

```
python
Operation
```

Step 3 Use the context manager to operate the file.

```
# Use the with statement to write file data.


# Use the with statement to read the file content.
```

Output

```
Python file operation
```

Step 4 Use the OS module to operate the file.

```
import os
# Rename the file.

# Delete the file.
```

2 Database Programming

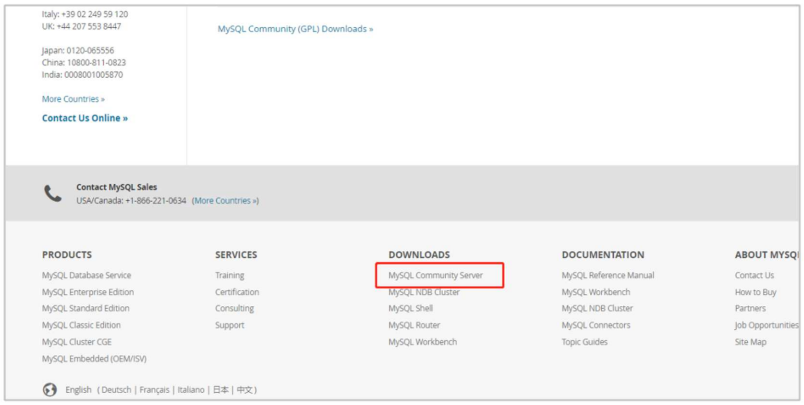
2.1 Database Programming

2.1.1 Installing the Database

Step 1 Download the installation package from the official website

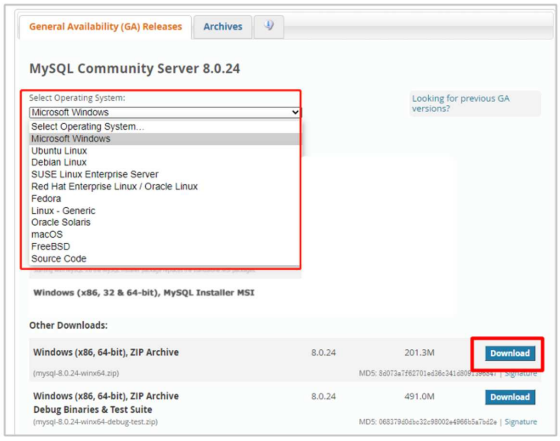
Login URL：https://www.mysql.com/

Choose DOWNLOADS to go to the Download page

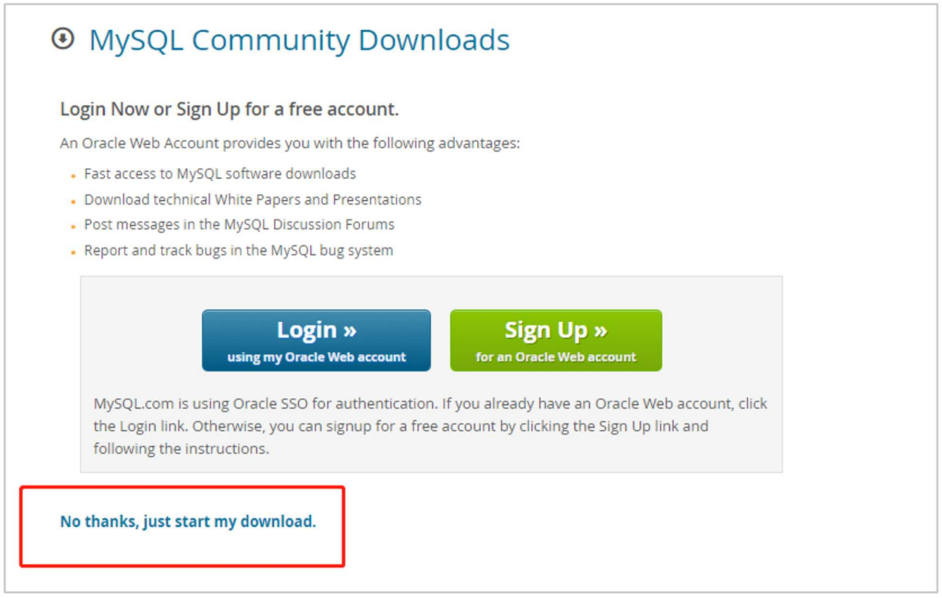


After that, select the MySQL Community Server at the bottom of the page

So here you can choose your own version (Windows,Linux,Mac) according to your operating system. Here we Windows as a display



Click Download after selecting the system



Here they want us to register an account, but we don't have to choose the following No thanks just start my download

After the download is complete, decompress the ZIP package to a directory.

Open the decompressed file, create the **my.ini** configuration file in the folder, and edit the **my.ini** configuration file to configure the following basic information:

```
[mysql]
# Set the default character set of the MySQL client.

[mysqld]
# Set the port 3306.

# Set the MySQL installation directory. Replace ## with the installation directory.
```

```
# Set the directory for storing the MySQL database data.
# datadir=##
# Maximum number of connections allowed.

# The default character set used by the server is the 8-bit latin1 character set.

# Default storage engine used when a new table is created.
```

Step 2

Step 3 Start MySQL.

Open the CLI as the administrator, switch to the decompressed folder, and go to the **bin** directory.

After initialization, you obtain a random password in the following format: root@localhost:*Random password*.

Run the following command to install the MySQL:

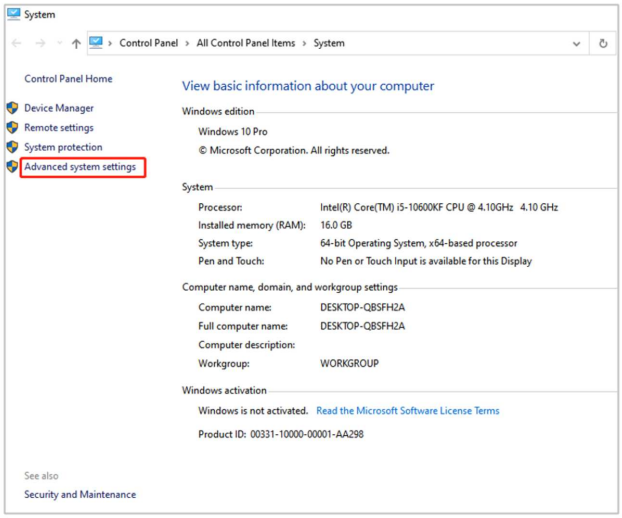
Start the MySQL database.

Log in to the MySQL database.

Change the password.

Step 4 Configure environment variables.

Right-click **My Computer**, choose **Properties** from the shortcut menu, and select **Advanced system settings**.



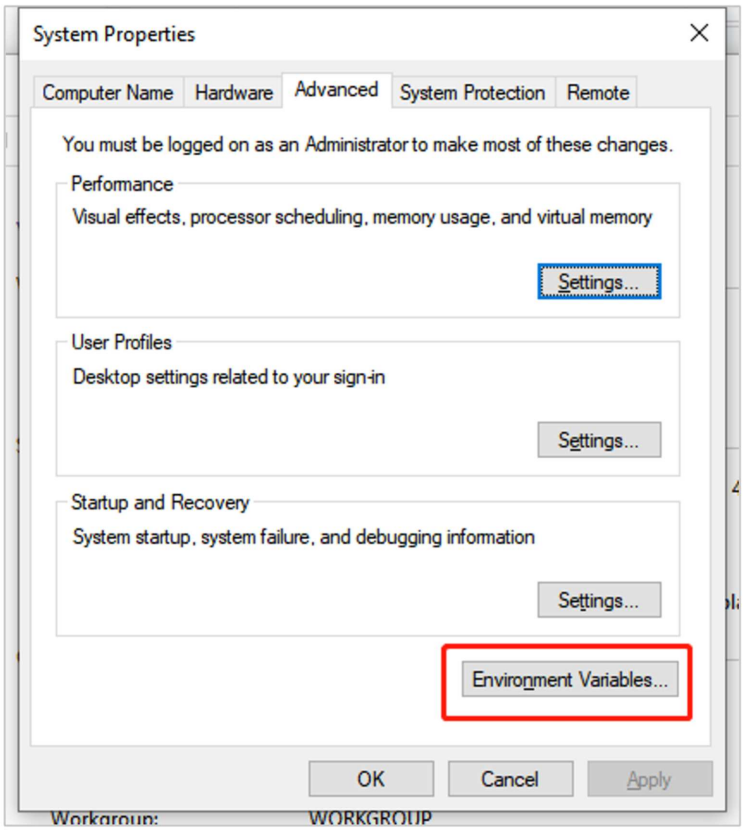
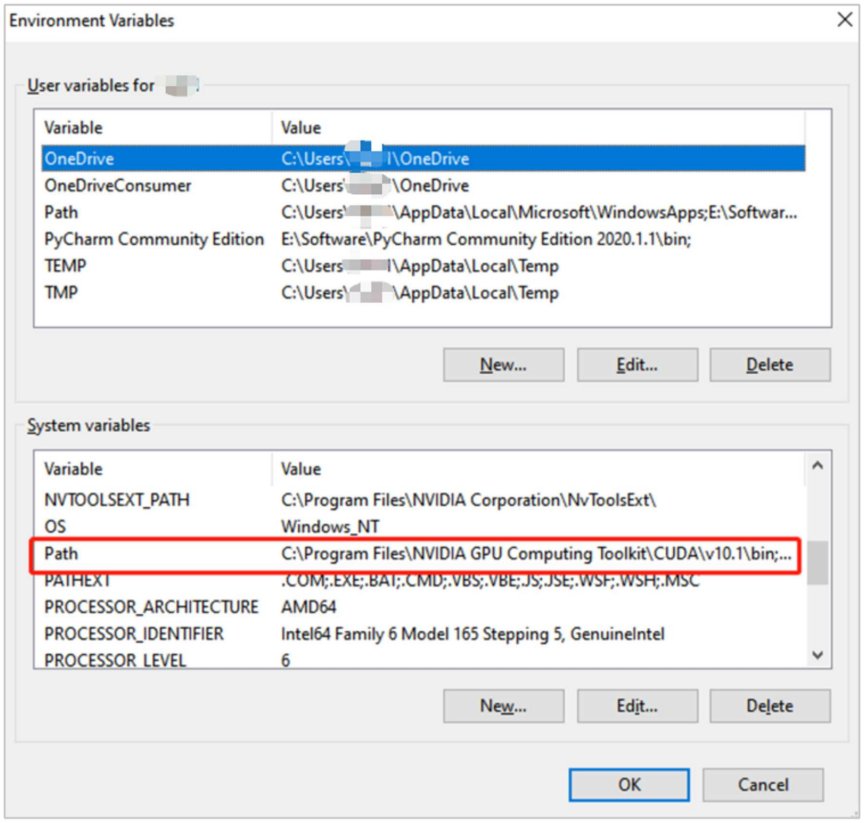


Figure 2-1 System Properties

Click **Environment Variables**. Add the database path to the **path** variable.


Environment Variables
User variables for
Variable
Value
OneDrive
C:\Users\ l\OneDrive
OneDriveConsumer
C:\Users\ \OneDrive
Path
C:\Users\ \AppData\Local\Microsoft\WindowsApps;E:\Softwar...
PyCharm Community Edition
E:\Software\PyCharm Community Edition 2020.1.1\bin;
TEMP
C:\Users l\AppData\Local\Temp
TMP
C:\Users\ \AppData\Local\Temp
New...
Edit...
Delete
System variables
Variable
Value
NVTOOLSEXT_PATH
C:\Program Files\NVIDIA Corporation\NvToolsExt\
OS
Windows_NT
Path
C:\Program Files\NVIDIA GPU Computing Toolkit\CUDA\v10.1\bin;...
PATHEXT
.COM;.EXE;.BAT;.CMD;.VBS;.VBE;.JS;.JSE;.WSF;.WSH;.MSC
PROCESSOR_ARCHITECTURE
AMD64
PROCESSOR_IDENTIFIER
Intel64 Family 6 Model 165 Stepping 5, GenuineIntel
PROCESSOR_LEVEL
6
New...
Edit...
Delete
OK
Cancel

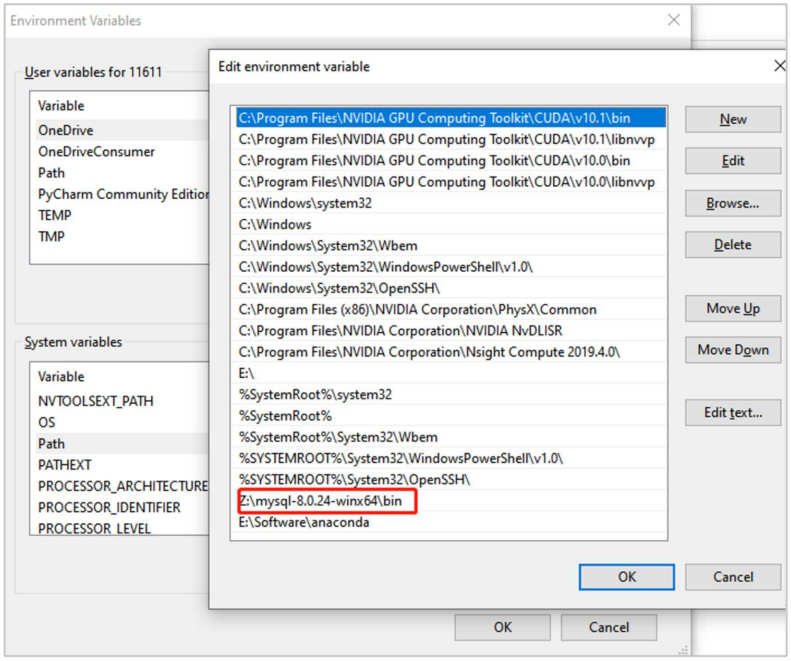

Environment Variables
User variables for 11611
Variable
OneDrive
OneDriveConsumer
Path
PyCharm Community Editior
TEMP
TMP
System variables
Variable
NVTOOLSEXT_PATH
OS
Path
PATHEXT
PROCESSOR_ARCHITECTURE
PROCESSOR_IDENTIFIER
PROCESSOR_LEVEL
OK
Cancel
Edit environment variable
C:\Program Files\NVIDIA GPU Computing Toolkit\CUDA\v10.1\bin
C:\Program Files\NVIDIA GPU Computing Toolkit\CUDA\v10.1\libnvvp
C:\Program Files\NVIDIA GPU Computing Toolkit\CUDA\v10.0\bin
C:\Program Files\NVIDIA GPU Computing Toolkit\CUDA\v10.0\libnvvp
C:\Windows\system32
C:\Windows
C:\Windows\System32\Wbem
C:\Windows\System32\WindowsPowerShell\v1.0\
C:\Windows\System32\OpenSSH\
C:\Program Files (x86)\NVIDIA Corporation\PhysX\Common
C:\Program Files\NVIDIA Corporation\NVIDIA NvDLISR
C:\Program Files\NVIDIA Corporation\Nsight Compute 2019.4.0\
E:\
%SystemRoot%\system32
%SystemRoot%
%SystemRoot%\System32\Wbem
%SYSTEMROOT%\System32\WindowsPowerShell\v1.0\
%SYSTEMROOT%\System32\OpenSSH\
Z:\mysql-8.0.24-winx64\bin
E:\Software\anaconda
New
Edit
Browse...
Delete
Move Up
Move Down
Edit text...
OK
Cancel

Figure 2-2 Configure environment variables.

2.2 Creating a database

Create database my_database.

```
mysql> create database my_database;
Query OK, 1 row affected (0.09 sec)
```

Figure 2-3 Creating a database

Switch the database.

```
mysql> use my_database;
Database changed
```

Figure 2-4 Switching the database

2.3 Database Programming

Step 1 Install modules required for the database.

Run the pip command to install a third-party library in Python.

Step 2 Connect to the database.

```
import pymysql
# Connect to the database.

# localhost: local connection. You can also change it to the IP address of the database.
# root: MySQL database account; mysql: database password.
# my_database: name of the database to be connected.
# Use the cursor() method to obtain the operation cursor.

# Execute the SQL statement using the execute method.

# Use the fetchone() method to obtain a data record.
```

Output:

```
Database version : 8.0.15
```

Step 3 Use Python to operate the database.

```
import pymysql
# Connect to the database.

# Use the cursor() method to obtain the operation cursor.

# SQL statement for creating a data table.



# Run the following commands:

# Close the database connection.
```

Check the result in the MySQL database.

```
ysql> show tables;
----------------------+
| Tables_in_my_database |
----------------------+
| my_table              |
----------------------+
row in set (0.00 sec)
```

Figure 2-5 Checking the MySQL database

Step 4 Data operations

Insert data.

```
import pymysql

# Connect to the database.

# Use the cursor() method to obtain the operation cursor.


# Execute the SQL INSERT statement.



# You only need to modify the SQL statement to be executed to delete or modify the data.

    # Execute the SQL statement.

    # Submit data to the database for execution.


    # Rollback upon an error.


# Close the database connection.
```

View the result in the database.

```
mysql> select * from my_table;
+------+------+
| id   | name |
+------+------+
|    1 | tom  |
+------+------+
1 row in set (0.00 sec)
```

Figure 2-6 Viewing the inserted data